Covering radii and deep-hole syndromes of length-eight Roth–Lempel codes

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Abstract

We determine covering radii and deep-hole syndromes for length-eight, dimension-four Roth–Lempel codes. They have the form $C = \mathrm{RL}_{4,\delta}(S)$, where $|S| = 6$. First, C is always MDS or NMDS, and it is NMDS precisely when δ is the sum of three distinct elements of S. Our main result is the sharp threshold $q \geq 16 \Rightarrow \rho(C) = 4$. For the prime-power fields below the threshold, the radius-three codes are classified completely up to affine equivalence: their numbers are respectively $2, 3, 14, 32, 7$ over $\mathbb{F}_7, \mathbb{F}_8, \mathbb{F}_9, \mathbb{F}_{11}, \mathbb{F}_{13}$, and no other radius-three case exists. The proof translates $\rho(C) \leq 3$ into a finite-geometric covering condition: the 56 planes spanned by triples of parity-check columns must cover $\mathrm{PG}(3, q)$. An uncovered point is both a deep-hole syndrome and a projection center producing a plane 8-arc. An explicit twisted-cubic construction proves the threshold theoretically for every $q \geq 23$, and for every NMDS member already when $q \geq 19$; within this theoretical proof, only thirty residual affine representatives over $q = 16, 17, 19$ require explicit finite certificates. We further give a 56-factor polynomial whose nonvanishing characterizes all deep-hole syndromes when $q \geq 16$, an explicit distance stratification in every admissible field, and closed relations among all projective syndrome shells, including the exact formula $m_2(C) = 28(q - 1) - 3r_S(\delta)$.

1 Introduction

Covering radii and deep holes of linear codes are classical questions in coding theory [2, 6, 11]. For MDS and NMDS codes, their syndrome geometry is naturally expressed by point configurations and arcs in finite projective spaces [1, 3]. Roth–Lempel codes are a classical source of non-Reed–Solomon MDS codes [12].

The recent literature closest to the present problem has developed in two directions. Han and Fan, and subsequently Xu and Zhou, investigate NMDS criteria and weight distributions for Roth–Lempel codes [5, 13]; Liang and Liao construct further NMDS classes from the Roth–Lempel matrix [9]. In parallel, covering radii and deep holes have become structural tools for extended twisted-GRS and non-GRS MDS–NMDS constructions [8, 7]. Distinct from these works, this paper determines the complete covering-radius threshold and explicit deep-hole syndrome criterion for the fixed length-eight, dimension-four Roth–Lempel family.

The problem is particularly sharp in this family. Since every $\mathrm{RL}_{4,\delta}(S)$ considered here is an $[8, 4]_q$ code, the redundancy bound gives $\rho(C) \leq 4$. Thus the only question is whether the covering radius drops from 4 to 3. We answer this question for every prime power $q \geq 7$, and at the same time determine the corresponding deep-hole syndromes.

The main results are as follows.

1. Every $\mathrm{RL}_{4,\delta}(S)$, $|S| = 6$, is MDS or NMDS. Moreover, it is NMDS if and only if $\delta = a+b+c$ for three distinct elements $a, b, c \in S$ (Proposition 2.1).

2. The covering-radius threshold is exact:

$$\rho\big(\mathrm{RL}_{4,\delta}(S)\big) = 4 \qquad \text{for every prime power } q \geq 16.$$

Below the threshold, all radius-three affine classes are determined: there are $2, 3, 14, 32, 7$ over $q = 7, 8, 9, 11, 13$, respectively (Theorems 4.1 and 4.2).

3. A syndrome has distance at most 3 exactly when its projective point lies in one of the 56 planes spanned by triples of parity-check columns (Proposition 3.1). Consequently, uncovered points are simultaneously deep-hole syndromes and projection centers that produce plane 8-arcs (Lemma 3.2).

4. For $q \geq 16$, all deep-hole syndromes are characterized by the nonvanishing of one explicit product of 56 linear factors (Proposition 3.3). The twisted-cubic subfamily of centers already proves existence for all $q \geq 23$, and for all NMDS codes when $q \geq 19$ (Proposition 3.4).

5. The complete syndrome distance stratification is explicit in every admissible field, and the projective shell sizes satisfy closed relations in terms of q, $r_S(\delta)$, and $m_4(C)$. In particular, the second shell is

$$m_2(C) = 28(q-1) - 3r_S(\delta),$$

where $r_S(\delta)$ counts the three-subsets of S with sum δ (Theorem 3.6 and Corollary 3.8).

Thus the paper gives a complete structure theorem for the classical Roth–Lempel family of MDS and NMDS codes. The result also supplies an infinite collection of certified $[8,4]_q$ Roth–Lempel seed codes with maximal covering radius and explicitly decidable deep-hole syndromes for deep-hole-based extension constructions; see Corollary 4.4. The finite part of the proof uses exact Magma enumeration, not randomized search. The proof-critical computations, certificates, and independent cross-checks are separated and documented in Section 8.

2 Roth–Lempel codes

Let $3 \leq k \leq n \leq q$, let $S = \{a_1, \dots, a_n\} \subseteq \mathbb{F}_q$, and let $\delta \in \mathbb{F}_q$. The Roth–Lempel code $\mathrm{RL}_{k,\delta}(S)$ is the $[n+2, k]_q$ code

$$\Big\{ \big(f(a_1), \dots, f(a_n), f_{k-1}, f_{k-2} + \delta f_{k-1}\big) : f(x) = \sum_{i=0}^{k-1} f_i x^i \in \mathbb{F}_q[x] \Big\}.$$

Equivalently, it has generator matrix

$$G_{\mathrm{RL}_{k,\delta}(S)} = \begin{pmatrix} 1 & 1 & \cdots & 1 & 0 & 0 \\ a_1 & a_2 & \cdots & a_n & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ a_1^{k-3} & a_2^{k-3} & \cdots & a_n^{k-3} & 0 & 0 \\ a_1^{k-2} & a_2^{k-2} & \cdots & a_n^{k-2} & 0 & 1 \\ a_1^{k-1} & a_2^{k-1} & \cdots & a_n^{k-1} & 1 & \delta \end{pmatrix}.$$

We focus on $k = 4$ and $n = 6$, so that the resulting code has parameters $[8,4]_q$. We use the standard convention that an $[N,K]_q$ code is NMDS when $d(C) = N - K$ and $d(C^\perp) = K$, following Dodunekov and Landgev [3].

The natural affine change of variables

$$x \mapsto ax + b, \qquad a \in \mathbb{F}_q^*,\ b \in \mathbb{F}_q,$$

acts on parameters by

$$S \mapsto aS + b, \qquad \delta \mapsto a\delta + 3b.$$

All classifications below are taken modulo this affine action.

Proposition 2.1. *For every prime power $q \geq 7$, every code $\mathrm{RL}_{4,\delta}(S)$, with $S \subseteq \mathbb{F}_q$ and $|S| = 6$, is either MDS or NMDS. More precisely, it is NMDS if and only if*

$$\delta = a + b + c \quad \textit{for some distinct } a, b, c \in S.$$

Proof. Write the last two generator columns as $e = (0,0,0,1)^\mathsf{T}$ and $f = (0,0,1,\delta)^\mathsf{T}$, and the six evaluation columns as $v(a) = (1, a, a^2, a^3)^\mathsf{T}$, $a \in S$. Every three columns are independent: this is the Vandermonde determinant for three evaluation columns, and with one or two of e, f it follows respectively from the rank-two projection of two distinct evaluation columns onto the first two coordinates, or from the nonzero first coordinate of a single evaluation column together with the independence of e, f. Every five columns have rank four. This is immediate if at least four evaluation columns occur; if exactly three occur, then both e, f occur and quotienting by their span leaves the rank-two matrix with columns $(1, a)^\mathsf{T}$ for three distinct a's. Thus either every four columns are independent, giving an MDS code, or some four are dependent while every three are independent and every five have rank four, giving an NMDS code. The only four-column sets that can become dependent are $\{v(a), v(b), v(c), f\}$: their determinant is a nonzero Vandermonde factor times $\delta - (a + b + c)$. This proves the final assertion. □

Lemma 2.2. *Put $\sigma = \sum_{a \in S} a$, $\theta = \sigma - \delta$, and*

$$\lambda_i = \left(\prod_{j \neq i} (a_i - a_j) \right)^{-1}.$$

For $C = \mathrm{RL}_{4,\delta}(S)$, a parity-check matrix is

$$H = \begin{pmatrix} \lambda_1 & \cdots & \lambda_6 & 0 & 0 \\ \lambda_1 a_1 & \cdots & \lambda_6 a_6 & 0 & 0 \\ \lambda_1 a_1^2 & \cdots & \lambda_6 a_6^2 & -1 & 0 \\ \lambda_1 a_1^3 & \cdots & \lambda_6 a_6^3 & \delta - \sigma & -1 \end{pmatrix}.$$

Consequently $C^\perp$ is monomially equivalent to $\mathrm{RL}_{4,\theta}(S)$.

Proof. The Lagrange denominator identities give

$$\sum_{i=1}^{6} \lambda_i a_i^r = 0 \quad (0 \leq r \leq 4), \qquad \sum_{i=1}^{6} \lambda_i a_i^5 = 1, \qquad \sum_{i=1}^{6} \lambda_i a_i^6 = \sigma.$$

A direct multiplication by $G_{\mathrm{RL}_{4,\delta}(S)}$ therefore gives $GH^\mathsf{T} = 0$, and H has rank 4. Scaling its first six columns by λ_i^{-1}, scaling the last two by -1, and swapping the last two columns produces the eight columns

$$(1, a, a^2, a^3)^\mathsf{T} \ (a \in S), \qquad (0,0,0,1)^\mathsf{T}, \qquad (0,0,1,\theta)^\mathsf{T},$$

which are precisely the generator columns of $\mathrm{RL}_{4,\theta}(S)$. □

Corollary 2.3. *Let $r_S(\delta)$ denote the number of three-subsets $T \subset S$ with $\sum_{a \in T} a = \delta$. Then C is MDS if and only if $r_S(\delta) = 0$, and it is NMDS if and only if $r_S(\delta) > 0$. Moreover,*

$$r_S(\delta) = r_S\left(\sum_{a \in S} a - \delta \right).$$

Proof. The type criterion is Proposition 2.1. Complementation maps a three-subset of sum δ bijectively to a three-subset of sum $\sum_{a \in S} a - \delta$, in agreement with Lemma 2.2. $\square$

Lemma 2.4. *For $C = \mathrm{RL}_{4,\delta}(S)$, let $r_S(\delta)$ be as in Corollary 2.3. The number $m_2(C)$ of projective syndrome points at distance exactly 2 satisfies*

$$m_2(C) = 28(q-1) - 3r_S(\delta).$$

Proof. The eight parity-check points determine 28 secant lines. Since every three columns are independent, two secants sharing a parity-check point have no further point in common. If no four columns are dependent, the union of the secants has

$$8 + 28(q-1)$$

points, and hence $m_2(C) = 28(q-1)$.

By Lemma 2.2, every dependent four-set of parity-check points has the form

$$\{V(a), V(b), V(c), E\}, \qquad a + b + c = \theta.$$

Equivalently, it is indexed by one of the $r_S(\delta) = r_S(\theta)$ complementary three-subsets. The three pairs of opposite secants in such a plane meet in three non-column points, decreasing the second shell by 3. Contributions from two different dependent four-sets cannot coincide. Each diagonal point lies on one secant through E. If it came from two circuits, this secant would be the same line $\langle E, V(a)\rangle$, and the opposite secants would meet at the same non-column point. If they share an endpoint, their intersection is a column; if they are disjoint, their four endpoints are coplanar, contrary to the Vandermonde determinant. Therefore the total decrement is exactly $3r_S(\delta)$. $\square$

3 A trisecant-plane criterion

Let C be an $[8,4]_q$ linear code and let

$$H_1, \dots, H_8 \in \mathrm{PG}(3,q)$$

be the projective columns of a parity-check matrix H of C. A syndrome $s \in \mathbb{F}_q^4$ has a coset representative of weight at most t if and only if its projective point lies in the projective span of at most t columns of H. For $t = 3$ this gives the following criterion.

Proposition 3.1. *Let C be an $[8,4]_q$ linear code with parity-check columns $H_1, \dots, H_8 \in \mathrm{PG}(3,q)$. Then*

$$\rho(C) \le 3 \iff \mathrm{PG}(3,q) = \bigcup_{1 \le i < j < \ell \le 8} \langle H_i, H_j, H_\ell \rangle.$$

Since $\rho(C) \le 4$ for every $[8,4]_q$ linear code, failure of this cover is equivalent to $\rho(C) = 4$.

Proof. The syndrome of a word supported on a set $I \subseteq \{1, \dots, 8\}$ is a linear combination of the columns $\{H_i : i \in I\}$. Thus a projective syndrome has a representative of weight at most 3 precisely when it lies in the span of one, two, or three parity-check columns. These smaller spans are contained in the spans of triples. Hence every syndrome has a representative of weight at most 3 if and only if the 56 triple spans cover $\mathrm{PG}(3,q)$. The final assertion is the standard redundancy bound [2, 6]. $\square$

Lemma 3.2. *With the notation of Proposition 3.1, a projective point $P \in \mathrm{PG}(3,q)$ is not covered by the 56 triple spans if and only if projection from P maps $H_1, \dots, H_8$ to eight points of a plane with no coincident pair and no collinear triple. In other words, uncovered syndrome points are exactly projection centers producing plane 8-arcs.*

Proof. Two projected columns coincide exactly when P lies on the line joining the corresponding two columns. Three projected columns are collinear exactly when P lies in the plane spanned by the corresponding three columns. Therefore absence from all triple spans is equivalent to the projected configuration having neither a collision nor a collinear triple. $\square$

Thus the uncovered syndrome points are not only deep-hole cosets; they are also constructive centers for producing plane 8-arcs from the Roth–Lempel parity-check configuration. The explicit centers constructed next therefore give a finite-geometric witness for the existence of such projected arcs in the non-exceptional range.

Proposition 3.3. *Let $C = \mathrm{RL}_{4,\delta}(S)$, let $\theta = \sum_{a\in S} a - \delta$, and choose homogeneous coordinates $P = (p_0, p_1, p_2, p_3)$ for a projective syndrome relative to the parity-check matrix in Lemma 2.2. Define*

$$\begin{aligned}
A_a(P) &= p_1 - ap_0,\\
B_{ab}(P) &= p_2 - (a+b)p_1 + abp_0,\\
C_{ab}(P) &= p_3 - \theta p_2 + \{ab + (a+b)(\theta - a - b)\}p_1 - ab(\theta - a - b)p_0,\\
D_{abc}(P) &= p_3 - (a+b+c)p_2 + (ab+ac+bc)p_1 - abcp_0
\end{aligned}$$

and

$$\Phi_{S,\delta}(P) = \prod_{a\in S} A_a(P) \prod_{\{a,b\}\subset S} B_{ab}(P)C_{ab}(P) \prod_{\{a,b,c\}\subset S} D_{abc}(P).$$

Then P is outside every triple span of parity-check columns if and only if $\Phi_{S,\delta}(P) \neq 0$. Equivalently, a syndrome has coset distance 4 if and only if its projective point satisfies this nonvanishing condition.

Proof. By Lemma 2.2, column scaling does not affect the projective spans, and the parity-check points may be written as

$$V(a) = (1, a, a^2, a^3)^\mathsf{T} \quad (a \in S), \qquad E = (0,0,1,\theta)^\mathsf{T}, \qquad F = (0,0,0,1)^\mathsf{T}.$$

The 56 triple spans split into four types. Their plane equations at P, up to nonzero scalar multiples, are respectively

triple type	number	plane equation
$\{V(a), E, F\}$	6	$A_a(P) = 0$
$\{V(a), V(b), F\}$	15	$B_{ab}(P) = 0$
$\{V(a), V(b), E\}$	15	$C_{ab}(P) = 0$
$\{V(a), V(b), V(c)\}$	20	$D_{abc}(P) = 0.$

For the third row, the cubic polynomial defining the plane has roots $a, b, \theta - a - b$. Hence P avoids all triple planes precisely when none of the displayed 56 factors vanishes. The final statement follows from Proposition 3.1. $\square$

Proposition 3.4. *Let $C = \mathrm{RL}_{4,\delta}(S)$, put $\theta = \sum_{a\in S} a - \delta$, and set*

$$\mathcal{B}_{S,\delta} = S \cup \{\theta - a - b : \{a,b\} \subset S\}.$$

For $x \in \mathbb{F}_q$, the point $V(x) = (1, x, x^2, x^3)$ has coset distance 4 if and only if $x \notin \mathcal{B}_{S,\delta}$. Let $m_4(C)$ denote the number of projective syndromes at distance 4, and let $r = r_S(\delta)$. Then

$$|\mathcal{B}_{S,\delta}| \leq 21 - 3r, \qquad m_4(C) \geq q - 21 + 3r.$$

Consequently:

$$\begin{array}{lcl}
q \geq 23 & \Longrightarrow & \rho(C) = 4, \quad m_4(C) \geq q - 21;\\
C \text{ NMDS and } q \geq 19 & \Longrightarrow & \rho(C) = 4, \quad m_4(C) \geq q - 18;\\
r \geq 2 \text{ and } q \geq 16 & \Longrightarrow & \rho(C) = 4, \quad m_4(C) \geq q - 15;\\
r \geq 3 \text{ and } q \geq 13 & \Longrightarrow & \rho(C) = 4, \quad m_4(C) \geq q - 12.
\end{array}$$

The corresponding lower bound for the number of deep-hole words is $q^4(q-1)(q-21+3r)$ whenever the displayed lower bound is positive.

Proof. Substitution of $P = V(x)$ into the factors of Proposition 3.3 gives

$$\begin{aligned} A_a(V(x)) &= x - a, \\ B_{ab}(V(x)) &= (x-a)(x-b), \\ C_{ab}(V(x)) &= (x-a)(x-b)(x-\theta+a+b), \\ D_{abc}(V(x)) &= (x-a)(x-b)(x-c). \end{aligned}$$

Hence $\Phi_{S,\delta}(V(x)) \neq 0$ precisely when $x \notin \mathcal{B}_{S,\delta}$.

By Corollary 2.3, exactly r three-subsets of S have sum θ. A pair $\{a,b\} \subset S$ satisfies $\theta - a - b \in S$ precisely when it lies in one of these three-subsets; each such subset supplies three pairs, and each qualifying pair has a unique third member. Thus exactly $3r$ of the 15 indexed pair values already belong to S. Any additional coincidence only decreases the forbidden set, giving

$$|\mathcal{B}_{S,\delta}| \leq 6 + (15 - 3r) = 21 - 3r.$$

There are therefore at least $q - 21 + 3r$ twisted-cubic centers whenever this number is positive. The projective-syndrome assertions follow from Proposition 3.3. Each projective syndrome coset contains $q^4(q-1)$ words, giving the stated lower bound for deep-hole words. □

Lemma 3.5. *Let $S' = uS + v$ and $\delta' = u\delta + 3v$, where $u \in \mathbb{F}_q^*$ and $v \in \mathbb{F}_q$. If $\theta = \sum_{a\in S} a - \delta$ and $\theta' = \sum_{a'\in S'} a' - \delta'$, then*

$$\theta' = u\theta + 3v, \qquad \mathcal{B}_{S',\delta'} = u\mathcal{B}_{S,\delta} + v.$$

In particular, the number of twisted-cubic centers supplied by Proposition 3.4 is invariant under the affine classification action.

Proof. Since $|S| = 6$,

$$\theta' = u\sum_{a\in S} a + 6v - u\delta - 3v = u\theta + 3v.$$

Furthermore,

$$\theta' - (ua+v) - (ub+v) = u(\theta - a - b) + v,$$

and the asserted transformation of $\mathcal{B}_{S,\delta}$ follows. □

Remark 3.1. For finite boundary certificates it is also useful to consider

$$W(x,\mu) = (1, x, x^2, x^3 + \mu), \qquad Q(t,u) = (0, 1, t, u).$$

Substitution in the factors of Proposition 3.3 shows that, once the first parameter avoids the corresponding secant obstructions, each remaining triple-plane obstruction excludes one value of the second parameter. These families compress the boundary certificates used later, but their elementary count does not improve the uniform threshold in Proposition 3.4.

Theorem 3.6. *Continue with the notation of Proposition 3.3, and let $P \neq 0$ be a syndrome point. Write $\mathcal{L}_2(P)$ for the following disjunction:*

$$\begin{array}{ll} p_0 = p_1 = 0; & \\ p_1 - ap_0 = p_2 - a^2p_0 = 0 & \text{for some } a \in S; \\ p_1 - ap_0 = 0, \quad p_3 - \theta p_2 - (a^3 - \theta a^2)p_0 = 0 & \text{for some } a \in S; \\ B_{ab}(P) = 0, \quad p_3 - (a+b)p_2 + abp_1 = 0 & \text{for some } \{a,b\} \subset S. \end{array}$$

Let $\mathcal{X} = \{V(a) : a \in S\} \cup \{E, F\}$, with $V(a), E, F$ as in the proof of Proposition 3.3. Then the coset distance represented by P is exactly

distance	*condition*
1	$P \in \mathcal{X}$,
2	$P \notin \mathcal{X}$ *and* $\mathcal{L}_2(P)$,
3	$\neg\mathcal{L}_2(P)$ *and* $\Phi_{S,\delta}(P) = 0$,
4	$\Phi_{S,\delta}(P) \neq 0$.

In particular, this gives an explicit deep-hole test also for every radius-three exceptional class in Theorem 4.1: its deep-hole syndromes are precisely the points satisfying $\neg\mathcal{L}_2(P)$.

Proof. The four lines in the definition of $\mathcal{L}_2(P)$ are respectively the equations for the 28 secant lines

$$\langle E, F\rangle, \quad \langle V(a), F\rangle, \quad \langle V(a), E\rangle, \quad \langle V(a), V(b)\rangle.$$

Thus $\mathcal{L}_2(P)$ is equivalent to syndrome distance at most 2. Proposition 3.3 says that $\Phi_{S,\delta}(P) = 0$ is equivalent to distance at most 3, while nonvanishing is equivalent to distance 4. Subtracting successive layers proves the displayed stratification. Finally, the trisecant cover is complete in a radius-three class, so only the first three layers remain. □

Lemma 3.7. *Let C be an $[8, 4]_q$ code and, for $1 \leq t \leq 4$, let $m_t(C)$ denote the number of projective syndrome points whose minimum expression as a span of parity-check columns uses exactly t columns. Then*

$$\rho(C) = \max\{t : m_t(C) > 0\}, \qquad |\mathrm{DH}(C)| = q^4(q-1)m_{\rho(C)}(C),$$

where $\mathrm{DH}(C)$ is the set of deep-hole words of C. In particular, if $\rho(C) = 4$, then $m_4(C)$ is exactly the number of uncovered points in Proposition 3.1.

Proof. A nonzero syndrome has coset distance t precisely when its projective point belongs to the t-th shell. Each projective point accounts for $q - 1$ nonzero scalar syndromes, and each syndrome coset contains $|C| = q^4$ words. The final assertion follows from the trisecant-plane criterion. □

Corollary 3.8. *Let $C = \mathrm{RL}_{4,\delta}(S)$, set $r = r_S(\delta)$, and write $h = m_4(C)$ for the number of uncovered projective syndrome points. Then*

$$\big(m_1(C), m_2(C), m_3(C), m_4(C)\big) = \big(8,\ 28(q-1) - 3r,\ q^3 + q^2 - 27q + 21 + 3r - h,\ h\big).$$

In particular, if $\rho(C) = 3$, then

$$|\mathrm{DH}(C)| = q^4(q-1)\big(q^3 + q^2 - 27q + 21 + 3r_S(\delta)\big).$$

Proof. The eight projective parity-check columns are distinct, so $m_1(C) = 8$. Lemma 2.4 gives $m_2(C)$, and

$$m_1(C) + m_2(C) + m_3(C) + m_4(C) = q^3 + q^2 + q + 1.$$

Solving for $m_3(C)$ gives the stated vector. If $\rho(C) = 3$, then $h = 0$, and Lemma 3.7 converts $m_3(C)$ to the number of deep-hole words. □

Corollary 3.9. *Let C be any $[8, 4]_q$ linear code. If $q \geq 56$, then $\rho(C) = 4$. In particular, for every prime power $q \geq 56$, every Roth–Lempel code $\mathrm{RL}_{4,\delta}(S)$ with $|S| = 6$ has covering radius 4.*

Proof. A plane in $\mathrm{PG}(3,q)$ contains q^2+q+1 points, while $\mathrm{PG}(3,q)$ contains q^3+q^2+q+1 points. The 56 planes spanned by triples of parity-check columns cover at most $56(q^2+q+1)$ points. For $q \geq 56$,

$$q^3+q^2+q+1 > 56(q^2+q+1).$$

Thus these planes cannot cover $\mathrm{PG}(3,q)$, so Proposition 3.1 gives $\rho(C) > 3$. Since an $[8,4]_q$ code has redundancy 4, the redundancy bound gives $\rho(C) \leq 4$. Therefore $\rho(C) = 4$. □

Proposition 3.10. *Let C be an $[8,4]_q$ code. If C is MDS and $q \geq 54$, then $\rho(C) = 4$. If C is NMDS and $q \geq 53$, then $\rho(C) = 4$.*

Proof. Assume first that C is MDS. Its 56 triple planes are distinct. At each of the eight columns, 21 triple planes meet, contributing at least 20 repeated incidences. Further, each of the 28 secant lines is contained in the six triple planes obtained by adjoining one of the remaining columns. The 28 sets of $q-1$ non-column points on these secants are pairwise disjoint: an intersection would put four columns in a plane. Hence the union of the triple planes has at most

$$56(q^2+q+1) - 8 \cdot 20 - 28(q-1) \cdot 5 = 56q^2 - 84q + 36$$

points. This is smaller than q^3+q^2+q+1 for $q \geq 54$.

If C is NMDS, some four columns are coplanar while every triple is independent. The four triple spans arising from that four-set coincide, so there are at most 53 distinct triple planes. Since

$$53(q^2+q+1) < q^3+q^2+q+1 \qquad (q \geq 53),$$

they do not cover $\mathrm{PG}(3,q)$. In either case, Proposition 3.1 and the redundancy bound yield $\rho(C) = 4$. □

4 Complete affine classifications over small fields

For each prime power

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53\}$$

all pairs (S, δ) with $|S| = 6$ were enumerated and reduced under the affine action from Section 2. For each affine representative, the covering radius was computed directly or radius 4 was certified by an explicit uncovered syndrome point for $\mathrm{RL}_{4,\delta}(S)$. This normalization is the natural affine specialization of the finite-geometric equivalence used for point configurations and arcs in projective spaces [1].

Theorem 4.1. *For the Roth–Lempel codes $\mathrm{RL}_{4,\delta}(S)$ with $|S| = 6$, the complete affine classifications over every prime-power field $\mathbb{F}_q$ with $7 \leq q \leq 53$ are summarized in Table 1. No class in these classifications has radius below 3.*

Table 1: Complete affine classifications for $\mathrm{RL}_{4,\delta}(S)$, $|S| = 6$, over $7 \leq q \leq 53$.

q	*raw cases*	*affine classes*	$\rho = 3$ *classes*	$\rho = 3$ *cases*	$\rho = 4$ *classes*	$\rho = 4$ *cases*
7	49	2	2	49	0	0
8	224	4	3	168	1	56
9	756	14	14	756	0	0
11	5082	48	32	3410	16	1672
13	22308	146	7	936	139	21372
16	128128	536	0	0	536	128128
17	210392	777	0	0	777	210392
19	515508	1514	0	0	1514	515508
23	2321781	4596	0	0	4596	2321781
25	4427500	7391	0	0	7391	4427500
27	7992270	11432	0	0	11432	7992270
29	13775580	16978	0	0	16978	13775580
31	22824711	24562	0	0	24562	22824711
32	28998144	29232	0	0	29232	28998144
37	86017008	64604	0	0	64604	86017008
41	184351908	112439	0	0	112439	184351908
43	262147522	145190	0	0	145190	262147522
47	504665931	233464	0	0	233464	504665931
49	685206984	291377	0	0	291377	685206984
53	1216746440	441540	0	0	441540	1216746440

In all twenty complete classifications, $\rho = 3$ holds if and only if the 56 trisecant planes cover $\mathrm{PG}(3, q)$.

Proof. The enumeration is finite and exact. For $q = 7, 8, 9$, every affine representative is constructed over its finite field and its covering radius is computed directly; this also verifies the assertion that no radius-two class occurs. For $q = 11, 13$, where radius-three classes persist, every affine representative is constructed in Magma and the trisecant-plane cover is computed directly; the covering radius computation was also run as an independent check. For $q = 16$, all 536 affine representatives are certified by the two-slice search, with full-space fallback for the residual classes. For $q = 17, 19, 23, 29, 31, 37$, where no hole-free class occurs, the scripts record an explicit uncovered projective syndrome point for every affine representative. The additional scans for $q = 25, 27, 32, 41, 43, 47, 49, 53$ first test one or two structured projective slices and record a point from a full-space fallback whenever both slices fail. The extension-field scans use field-element canonical representatives, not integer encodings. No class remains uncertified. Proposition 3.1 then gives $\rho = 4$ for every certified representative. Reproducibility files are listed in Section 8. $\square$

Theorem 4.2. *For every prime power $q \geq 16$, every Roth–Lempel code $\mathrm{RL}_{4,\delta}(S)$ with $|S| = 6$ has covering radius 4. For $7 \leq q < 16$, the complete radius-three class counts are those in Theorem 4.1: respectively $2, 3, 14, 32, 7$ for $q = 7, 8, 9, 11, 13$.*

Proof. For $q = 16, 17, 19$, Proposition 3.4 and direct evaluation of its one-parameter forbidden set leave only $12, 15, 3$ affine representatives, respectively. They are certified by the explicit families in Remark 3.1; the thirty certificates are listed in the companion supplement. Proposition 3.4 proves the claim theoretically for every $q \geq 23$, which covers every remaining prime power at least 16. The final assertion is the first five rows of Theorem 4.1. $\square$

Thus the boundary fields $q = 16, 17, 19$ are covered in two independent ways: the proof of Theorem 4.2 uses the twisted-cubic reduction plus thirty explicit residual certificates, while the complete enumeration in Theorem 4.1 supplies an independent class-by-class computational verification.

Corollary 4.3. *Let $q \geq 16$, let $C = \mathrm{RL}_{4,\delta}(S)$ with $|S| = 6$, and let u be a word with nonzero syndrome $P = Hu^{\mathsf{T}}$, using the explicit matrix H in Lemma 2.2. Then u is a deep hole of C if and only if*

$$\Phi_{S,\delta}(P) \neq 0.$$

Proof. Theorem 4.2 gives $\rho(C) = 4$ for every $q \geq 16$. Proposition 3.3 characterizes exactly those syndrome cosets whose distance is 4. $\square$

Corollary 4.4. *For every prime power $q \geq 16$, the length-eight Roth–Lempel family* $\mathrm{RL}_{4,\delta}(S)$*, $|S| = 6$, supplies explicit MDS or NMDS $[8,4]_q$ seed codes with maximal covering radius. More precisely, for each seed the deep-hole syndrome cosets are exactly the nonzero syndromes P satisfying $\Phi_{S,\delta}(P) \neq 0$. Therefore any deep-hole-based extension framework whose hypotheses require an explicit deep hole in such a seed, such as the framework in [7], may choose P by this nonvanishing test.*

Proof. This is an immediate restatement of Theorem 4.2 and Corollary 4.3. The MDS/NMDS alternative is Proposition 2.1. □

The three smallest fields have particularly compact exceptional descriptions. Here $\mathbb{F}_8 = \mathbb{F}_2(w)$, $w^3 + w + 1 = 0$, and $\mathbb{F}_9 = \mathbb{F}_3(w)$, $w^2 + 2w + 2 = 0$. The compact representatives are listed in Table 2; all displayed classes have $\rho = 3$.

Table 2: Compact representatives for the radius-three classes over $q = 7, 8, 9$.

q	type	S	δ	number of classes
7	NMDS	$\{0,1,2,3,4,5\}$	$0,4$	2
8	NMDS	$\{0,1,w,w^2,w^3,w^4\}$	$0,1,w^2$	3
9	NMDS	$\{0,1,w,w^2,w^3,2\}$	$\delta \in \mathbb{F}_9$	9
9	NMDS	$\{0,1,w,w^2,2,w^7\}$	$0,w,w^2,w^3$	4
9	MDS	$\{0,1,w,w^2,2,w^7\}$	1	1

The unique $\mathbb{F}_8$ class with $\rho = 4$ has the same displayed set $S = \{0,1,w,w^2,w^3,w^4\}$ and $\delta = w^5$. Together with Table 1, this shows that every affine class over $q = 7$ and $q = 9$ has radius 3, whereas radius-four classes already occur over $q = 8$ and $q = 11$.

The multiplicity $r_S(\delta)$ from Corollary 2.3 refines the MDS/NMDS labels. Its complete low-field radius spectrum is given in Table 3.

Table 3: Low-field radius spectrum by the multiplicity $r_S(\delta)$.

q	$r_S(\delta)$	ρ	classes	q	$r_S(\delta)$	ρ	classes
7	2	3	1	11	0	4	1
7	3	3	1	11	1	3	14
8	2	3	3	11	2	3	16
8	4	4	1	11	2	4	9
9	0	3	1	11	3	3	1
9	2	3	8	11	3	4	6
9	3	3	5	13	0	3	3
11	0	3	1	13	0	4	7
				13	1	3	4
				13	1	4	56
				13	2	4	64
				13	3	4	12

Thus $r_S(\delta) = 0$ distinguishes MDS from NMDS exactly, but neither the code type nor the number of dependent four-column sets determines the covering radius. For instance, over $\mathbb{F}_{11}$ both radius values occur when $r_S(\delta) = 0, 2$, or 3; over $\mathbb{F}_{13}$, all radius-three classes have $r_S(\delta) \leq 1$. In contrast, Lemma 2.4 shows that this same additive statistic determines the entire second syndrome shell exactly.

Lemma 3.7 also turns the exceptional classification into an exact deep-hole census. Table 4 gives the complete spectrum for the classes with $\rho = 3$; "DH cosets" denotes $(q-1)m_3(C)$, and every such coset contains q^4 deep-hole words.

Table 4: Deep-hole coset spectrum for the radius-three classes.

q	type	DH cosets	classes	raw	q	type	DH cosets	classes	raw
7	NMDS	1380	1	7	11	MDS	11760	1	55
7	NMDS	1398	1	42	11	NMDS	11790	14	1540
8	NMDS	2709	3	168	11	NMDS	11820	16	1705
9	MDS	4704	1	24	11	NMDS	11850	1	110
9	NMDS	4752	8	516	13	MDS	24432	3	312
9	NMDS	4776	5	216	13	NMDS	24468	4	624

For the radius-four classes, the hole distributions below constitute an equally complete deep-hole census through the formula

$$|\mathrm{DH}(C)| = q^4(q-1) \cdot \mathrm{holes}(C).$$

For example, the unique radius-four class over $\mathbb{F}_8$ has one hole and therefore $8^4 \cdot 7 = 28672$ deep-hole words.

For $q = 11$ and $q = 13$, where hole-free classes occur, the distribution by code type and by the number of uncovered projective syndrome points is shown in Table 5.

Table 5: Hole distribution for $q = 11$ and $q = 13$.

$q = 11$				$q = 13$			
type	holes	classes	raw	type	holes	classes	raw
MDS	0	1	55	MDS	0	3	312
NMDS	0	31	3355	NMDS	0	4	624
MDS	1	1	22	MDS	1	4	624
NMDS	1	13	1430	MDS	2	1	156
NMDS	2	2	220	MDS	3	2	312
				NMDS	1	11	1716
				NMDS	2	34	5304
				NMDS	3	22	3328
				NMDS	4	27	4082
				NMDS	5	15	2340
				NMDS	6	10	1560
				NMDS	7	8	1248
				NMDS	8	2	234
				NMDS	9	2	312
				NMDS	12	1	156

Here holes means the number of projective syndrome points not covered by the 56 trisecant planes. Thus the rows with holes 0 are precisely the classes with $\rho = 3$, and every remaining row has $\rho = 4$.

For $q = 17, 19, 23, 29, 31, 37$, Table 6 records the MDS/NMDS split in fields where no radius-three class occurs.

Table 6: MDS/NMDS classification for fields with no radius-three classes.

q	ρ	MDS classes	MDS raw	NMDS classes	NMDS raw
17	4	139	37128	638	173264
19	4	350	119434	1164	396074
23	4	1488	752268	3108	1569513
29	4	7326	5947514	9652	7828066
31	4	11316	10518771	13246	12305940
37	4	34472	45890064	30132	40126944

No affine class is hole-free for $q = 17, 19, 23, 29, 31, 37$. For $q = 17$, where full hole counts were recorded, the number of uncovered projective syndrome points ranges from 28 to 117 across the 777 classes. For $q = 19, 23, 29, 31, 37$, the original computation records an explicit first uncovered syndrome point for every affine class.

An earlier one-slice certificate computation, which searches first in the structured projective slice $P = [1, u, v, v]$, is retained in the reproducibility manifest as a historical cross-check. The stronger two-slice scan is the compact form reported in the article.

A strengthened two-slice scan first searches $P = [1, u, v, v]$, then $P = [1, u, v, u]$, and invokes full-space search only if both slices fail. It handles both prime and non-prime fields. As shown in Table 7, the final column vanishes in every field listed.

Table 7: Two-slice certificate search for prime and extension fields.

q	affine classes	$[1, u, v, v]$ cert.	$[1, u, v, u]$ cert.	fallback cert.	uncertified
16	536	435	74	27	0
17	777	695	71	11	0
25	7391	6787	568	36	0
27	11432	10563	822	47	0
32	29232	27259	1866	107	0
43	145190	138770	6128	292	0
47	233464	224118	8966	380	0
49	291377	279078	11914	385	0
53	441540	425601	15368	571	0

The $\mathbb{F}_{11}$ classes with $\rho = 3$ can be summarized compactly by affine representative families, as in Table 8.

Table 8: Compact representative families for the $q = 11$ radius-three classes.

type	S	δ
MDS	$\{0, 1, 2, 3, 4, 10\}$	10
NMDS	$\{0, 1, 2, 3, 4, 10\}$	$0, 1$
NMDS	$\{0, 1, 2, 3, 4, 6\}$	$1, 2, 3, 4, 5, 10$
NMDS	$\{0, 1, 2, 3, 4, 7\}$	$1, 2, 3, 4, 5, 6, 7, 8, 9, 10$
NMDS	$\{0, 1, 2, 3, 5, 6\}$	$1, 2, 4, 5, 6, 7$
NMDS	$\{0, 1, 2, 3, 5, 9\}$	$0, 1, 2, 3, 4, 10$
NMDS	$\{0, 1, 2, 4, 5, 7\}$	2

These rows account for 32 affine classes; the full $q = 11$ affine list is given in Section 5.

Corollary 4.5. *Over $\mathbb{F}_{13}$, exactly the seven affine classes in Table 9 have covering radius* 3.

Table 9: The seven radius-three affine classes over $\mathbb{F}_{13}$.

type	S	δ	*orbit size*
MDS	$\{0, 1, 2, 10, 11, 12\}$	5	78
MDS	$\{0, 1, 2, 3, 10, 12\}$	7	156
MDS	$\{0, 1, 2, 3, 5, 11\}$	11	78
NMDS	$\{0, 1, 2, 3, 4, 8\}$	10	156
NMDS	$\{0, 1, 2, 3, 5, 7\}$	0	156
NMDS	$\{0, 1, 2, 3, 5, 7\}$	5	156
NMDS	$\{0, 1, 2, 3, 5, 9\}$	11	156

All remaining 139 *affine classes have covering radius* 4.

5 Complete $q = 11$ affine classification

This section records the complete affine classification for $\mathrm{RL}_{4,\delta}(S)$ over $\mathbb{F}_{11}$, with $|S| = 6$. There are 48 affine classes representing 5082 raw parameter pairs (S, δ). Exactly 32 affine classes, representing 3410 raw pairs, have $\rho = 3$; the remaining 16 affine classes, representing 1672 raw pairs, have $\rho = 4$. The complete list is given in Table 11, and Table 10 summarizes the same classification by type, radius, and number of holes.

Table 10: Summary of the $q = 11$ affine classification by type, radius, and holes.

type	ρ	holes	affine classes	raw cases
MDS	3	0	1	55
NMDS	3	0	31	3355
MDS	4	1	1	22
NMDS	4	1	13	1430
NMDS	4	2	2	220

Here holes denotes the number of projective syndrome points not covered by the 56 trisecant planes. Thus the hole-free rows are exactly the $\rho = 3$ rows.

The set labels used in Table 11 are

$$\begin{aligned} S_1 &= \{0,1,2,3,5,6\}, & S_2 &= \{0,1,2,3,4,7\}, \\ S_3 &= \{0,1,2,3,4,6\}, & S_4 &= \{0,1,2,3,4,10\}, \\ S_5 &= \{0,1,2,3,5,9\}, & S_6 &= \{0,1,2,4,5,7\}. \end{aligned}$$

Table 11: Complete affine classification over $\mathbb{F}_{11}$ for $k = 4, n = 6$.

idx	orb	type	ρ	S	δ	holes	idx	orb	type	ρ	S	δ	holes
1	110	NMDS	3	S_1	7	0	25	110	NMDS	3	S_3	2	0
2	110	NMDS	3	S_1	5	0	26	110	NMDS	4	S_3	0	1
3	110	NMDS	4	S_1	3	1	27	110	NMDS	4	S_3	9	1
4	110	NMDS	3	S_1	1	0	28	110	NMDS	4	S_3	7	1
5	110	NMDS	4	S_1	10	1	29	110	NMDS	3	S_3	5	0
6	110	NMDS	4	S_1	8	1	30	110	NMDS	3	S_3	3	0
7	110	NMDS	3	S_1	6	0	31	110	NMDS	3	S_3	1	0
8	110	NMDS	3	S_1	4	0	32	110	NMDS	3	S_3	10	0
9	110	NMDS	3	S_1	2	0	33	110	NMDS	4	S_3	8	1
10	110	NMDS	4	S_1	0	1	34	110	NMDS	4	S_4	4	2
11	110	NMDS	4	S_1	9	1	35	110	NMDS	3	S_4	0	0
12	110	NMDS	3	S_2	9	0	36	110	NMDS	4	S_4	2	1
13	110	NMDS	3	S_2	1	0	37	110	NMDS	4	S_4	3	2
14	110	NMDS	3	S_2	4	0	38	55	MDS	3	S_4	10	0
15	110	NMDS	3	S_2	7	0	39	110	NMDS	3	S_4	1	0
16	110	NMDS	3	S_2	10	0	40	110	NMDS	3	S_5	2	0
17	110	NMDS	3	S_2	2	0	41	110	NMDS	3	S_5	1	0
18	110	NMDS	3	S_2	5	0	42	110	NMDS	3	S_5	0	0
19	110	NMDS	3	S_2	8	0	43	55	NMDS	3	S_5	10	0
20	110	NMDS	4	S_2	0	1	44	110	NMDS	3	S_5	3	0
21	110	NMDS	3	S_2	3	0	45	110	NMDS	3	S_5	4	0
22	110	NMDS	3	S_2	6	0	46	110	NMDS	4	S_6	0	1
23	110	NMDS	4	S_3	6	1	47	110	NMDS	3	S_6	2	0
24	110	NMDS	3	S_3	4	0	48	22	MDS	4	S_6	4	1

Here "idx" and "orb" denote the class index and affine-orbit size, respectively.

Companion supplementary tables.

Complete affine-class tables for

$$q \in \{13, 17, 19, 23, 29, 31, 37\}$$

are supplied in the companion supplementary file in the public repository. For $q = 19, 23, 29, 31, 37$, each listed class includes an explicit first-hole certificate. The full class-level results for $q = 7, 8, 9$ are contained in the exact small-prime-power log. Exact certificate logs for the remaining certified fields are listed in Section 8 and are available in the public repository cited there.

6 Discussion

The classification separates two related but distinct effects. The MDS or NMDS status of a Roth–Lempel code is governed by additive constraints on subsets of S. The covering-radius drop from 4 to 3, however, is a global covering property of the eight parity-check points in $\mathrm{PG}(3, q)$. In the language of Lemma 3.2, the classes with $\rho = 4$ admit at least one projection center producing a plane 8-arc; the exceptional classes with $\rho = 3$ admit no such center.

This viewpoint suggests a finite-geometric version of the general problem: classify eight-point configurations in $\mathrm{PG}(3, q)$ arising from $\mathrm{RL}_{4,\delta}(S)$ whose 56 trisecant planes cover the ambient space. The complete classifications show that this cover property occurs throughout $q = 7, 9$, in three of four classes at $q = 8$, is common at $q = 11$, rare at $q = 13$, and absent in every complete affine classification over prime-power fields $16 \le q \le 53$; the seven exceptional classes over $\mathbb{F}_{13}$ are listed in Corollary 4.5. Corollary 3.9 makes this rigorous for arbitrary $[8, 4]_q$ codes once $q \ge 56$, while Proposition 3.10 lowers the relevant thresholds to $q \ge 54$ for MDS codes and $q \ge 53$ for NMDS codes. More sharply for the Roth–Lempel family itself, Proposition 3.4 replaces this tail argument by explicit deep-hole centers for every $q \ge 23$, and for every NMDS member already when $q \ge 19$. Consequently Theorem 4.2 is a complete statement for the entire length-eight family: a covering-radius drop can occur only below $q = 16$, and it occurs exactly in the classes enumerated over $q = 7, 8, 9, 11, 13$. Proposition 3.3 and Corollary 4.3 add a complementary infinite conclusion: once $q \ge 16$, the deep holes are not merely known to exist; their syndrome cosets are given by one explicit nonvanishing condition in four variables. Theorem 3.6 removes the remaining descriptive gap below this threshold: in the exceptional radius-three classes, the deep holes are given explicitly by exclusion from the 28 displayed secant lines. Finally, the $r_S(\delta)$-spectrum shows why MDS/NMDS status alone cannot replace the trisecant-cover condition, even though Lemma 2.4 shows that it completely controls the second shell. Lemma 3.5 also ensures that the new twisted-cubic certificates are intrinsic to affine classes rather than artifacts of the chosen representatives. Corollary 3.8 further replaces the radius-three deep-hole census by a closed formula depending only on q and $r_S(\delta)$. From the matroidal viewpoint on linear codes, where the Tutte polynomial records weight-enumerator information [4], the exact formula for $m_2(C)$ suggests that these length-eight Roth–Lempel column matroids form a particularly rigid additive subfamily.

For reference, Table 12 gives the exact evaluation of the displayed certificate families on the three boundary fields and on $q = 23$, the first field covered uniformly by the twisted-cubic criterion.

Table 12: Boundary-field certificate counts for the displayed center families.

q	affine classes	$V(x)$ certified	$V(x) + \mu F$ additional	$Q(t, u)$ residual
16	536	524	11	1
17	777	762	15	0
19	1514	1511	3	0
23	4596	4596	0	0

Here $Q(t, u) = (0, 1, t, u)$. The sole final residual before this third family is the MDS class over $\mathbb{F}_{16}$ with

$$S = \{1, w^{12}, w, w^{14}, 0, w^8\}, \qquad \delta = w, \qquad Q(0, w^2) = (0, 1, 0, w^2).$$

These two-parameter families are used only as explicit finite certificates; no stronger uniform threshold theorem is asserted for them.

7 Conclusion

For the length-eight Roth–Lempel family $\mathrm{RL}_{4,\delta}(S)$, $|S| = 6$, the covering-radius problem is completely settled. Every member is MDS or NMDS, the exact threshold is $\rho(C) = 4$ for all prime powers $q \geq 16$, and the only radius-three classes occur over $q = 7, 8, 9, 11, 13$, where they are all explicitly classified up to affine equivalence. Moreover, the deep-hole syndromes are described by an explicit 56-factor condition in the maximal-radius range, while the exceptional low-field cases are covered by the explicit syndrome stratification and shell formulas. Thus the paper gives a complete answer for this $[8, 4]$ Roth–Lempel family and provides certified seed codes for possible deep-hole-based extension constructions; the broader problem for arbitrary Roth–Lempel parameters (k, n) remains separate. Extending the same syndrome-geometric method beyond length eight is a natural direction for future work.

8 Reproducibility

All finite computations reported in this paper were performed in Magma [10], using exact finite-field arithmetic. Parameter pairs are reduced under the affine action

$$(S, \delta) \longmapsto (uS + v, u\delta + 3v), \qquad u \in \mathbb{F}_q^*, \quad v \in \mathbb{F}_q.$$

No randomized search is used for a classification or for a certificate entering a theorem.

Proof-critical computations

Only two finite verification tasks enter the main results. First, for $q = 7, 8, 9, 11, 13$, exhaustive enumeration of affine representatives gives the exceptional-field covering radii, distance shells, and deep-hole spectra used in the complete small-field classification. Second, in the three boundary fields $q = 16, 17, 19$, the theoretical criterion of Proposition 3.4 resolves every representative except $12, 15, 3$, respectively. The remaining thirty representatives are settled by explicit centers from the families $W(x, \mu)$ and $Q(t, u)$ in Remark 3.1; the certificates themselves are listed in the companion supplement available in the public repository. For every $q \geq 23$, Theorem 4.2 is theoretical and does not depend on an exhaustive machine scan. The complete-classification scans reported in Theorem 4.1, including the $q = 16$ two-slice scan, provide independent class-by-class checks of these conclusions rather than additional hypotheses for the threshold proof.

Verification summary

Table 13 summarizes the verification tasks used by the proof and separates proof-critical computations from cross-checks.

Table 13: Verification tasks and their logical role.

Task	Fields	Certified output
Exceptional classification	$7, 8, 9, 11, 13$	All affine classes; covering radius; complete distance shells and deep-hole spectra.
Boundary certification	$16, 17, 19$	All residual cases after the theoretical test; thirty explicit radius-four witnesses.
Boundary class-level cross-checks	$16, 17, 19$	Complete affine-representative scans, including the $q = 16$ two-slice scan and explicit uncovered syndromes for $q = 17, 19$.
Formula validation	$7, \ldots, 19$	Product criterion, shell formula, and affine-covariance checks.
Independent cross-checks	$23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53$	Additional exact scans consistent with the theorem; not proof inputs.

The accompanying program-and-log reproducibility manifest records the Magma files, exact output logs, extracted tables, and the logical role of each computation. This keeps the article-level argument separate from archival computational material while providing a direct audit trail for every finite assertion used above.

Data and code availability

The Magma programs, output logs, boundary certificates, extracted data tables, manuscript source, and reproducibility manifest are publicly available at

`https://github.com/matheorist/roth-lempel-covering-radius`.

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